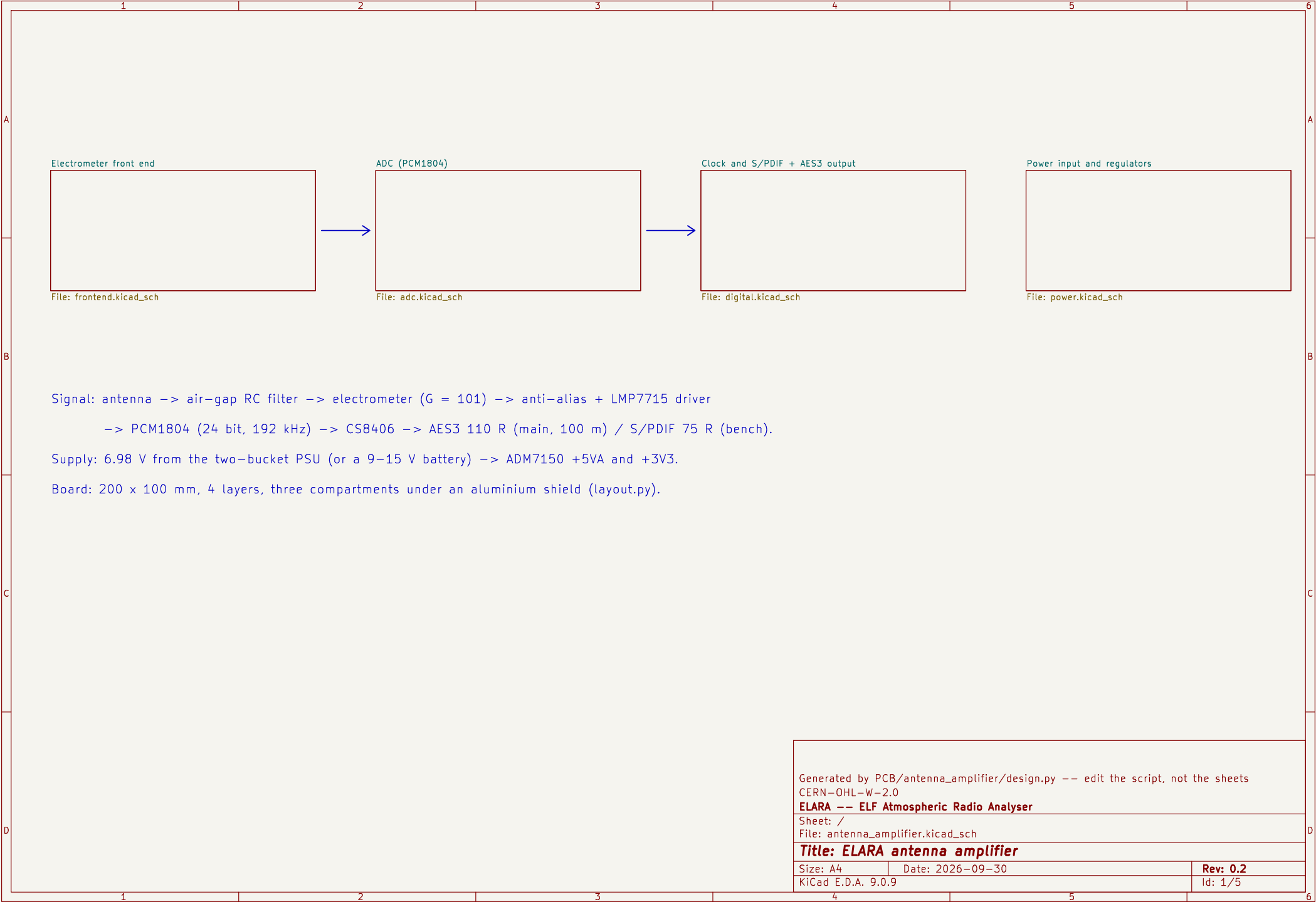
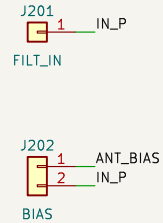


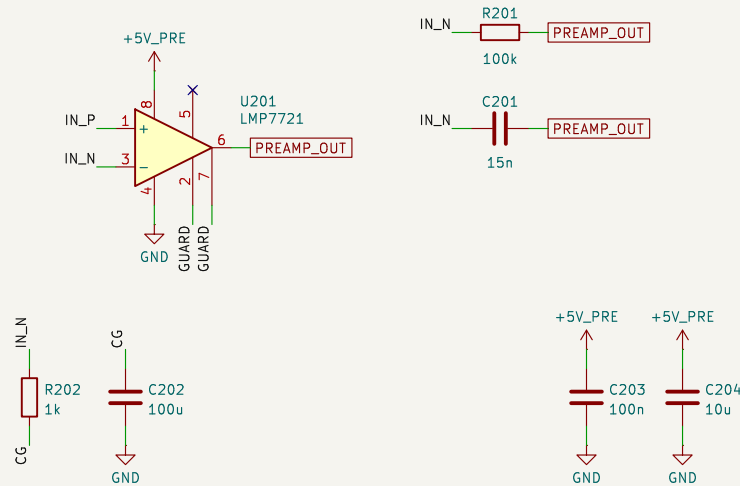


INPUT (comp. 1)

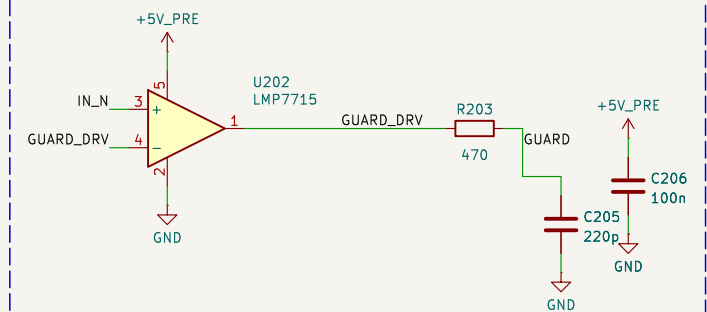


IN_P: guard on all layers,
no mask, PTFE turret.
J202: fit the G-ohm bias R.

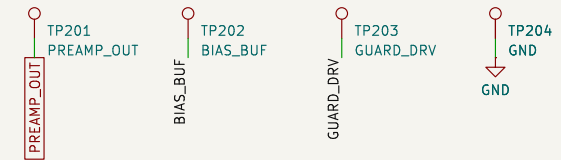
ELECTROMETER $G = 101$



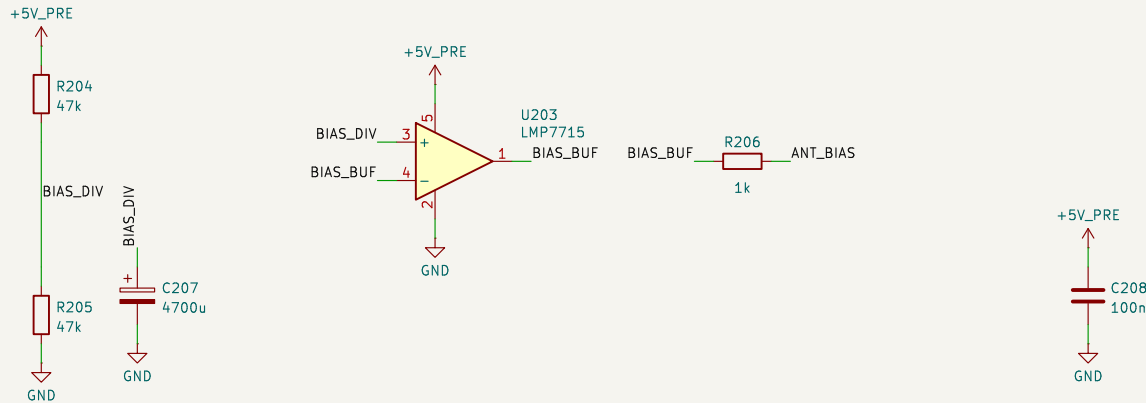
GUARD DRIVER



TEST POINTS



ANTENNA BIAS 2.5 V



Cg returns to GND: the 2.5 V buffer noise stays out of the gain path.
Guard buffer senses IN- (virtual short to IN+): no bias current on IN_P.

Generated by PCB/antenna_amplifier/design.py -- edit the script, not the sheets
CERN-OHL-W-2.0

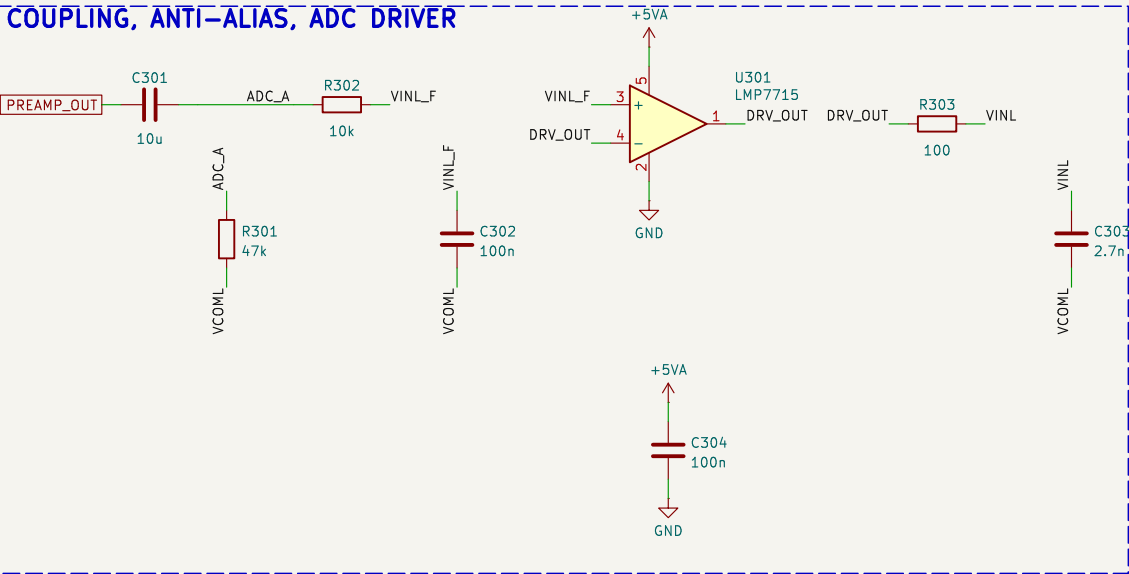
ELARA -- ELF Atmospheric Radio Analyser

Sheet: /Electrometer front end/
File: frontend.kicad_sch

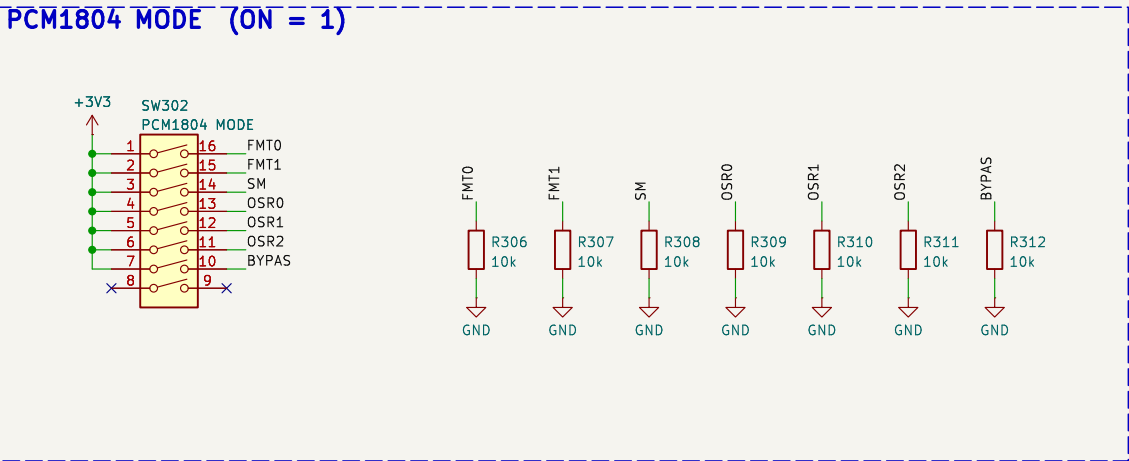
Title: ELARA antenna amplifier -- Electrometer front end

Size: A4	Date: 2026-09-30	Rev: 0.2
KiCad E.D.A. 9.0.9		Id: 2/5

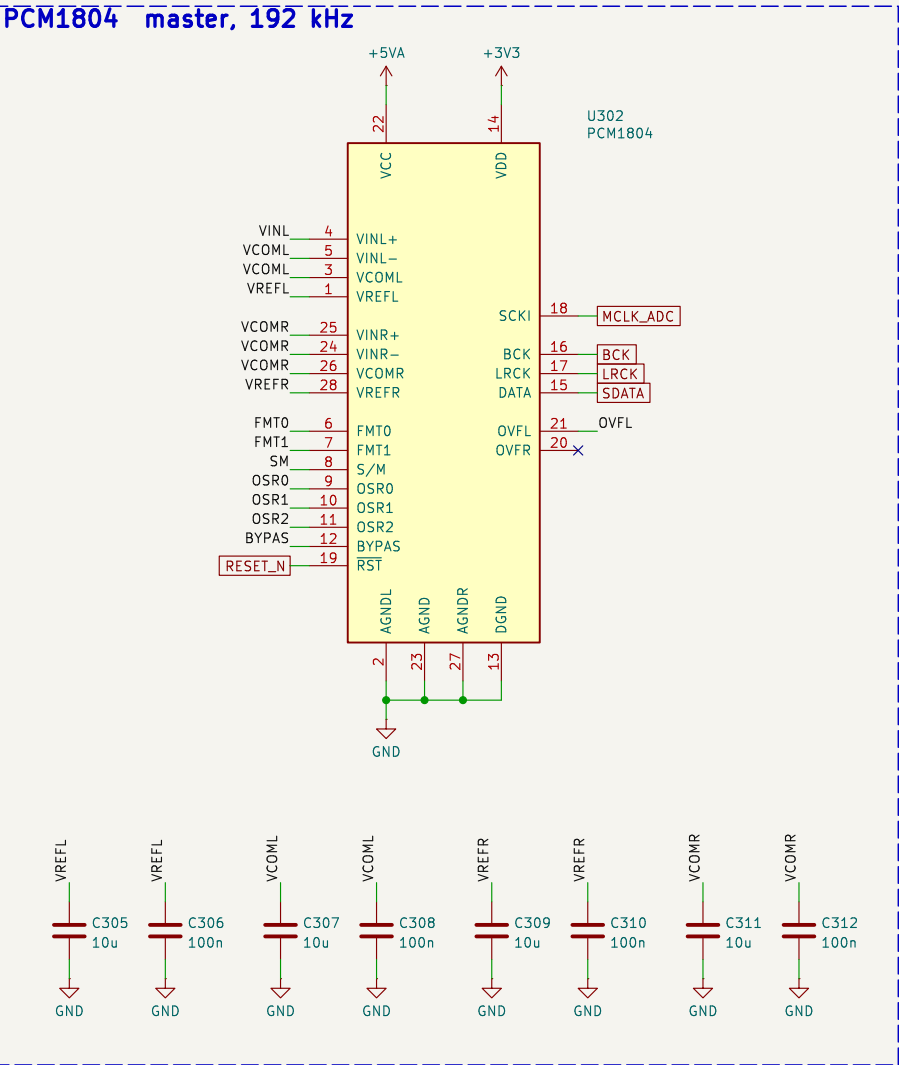
COUPLING, ANTI-ALIAS, ADC DRIVER



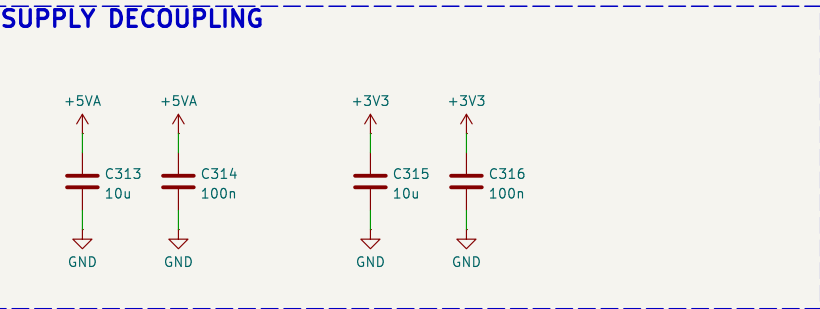
PCM1804 MODE (ON = 1)



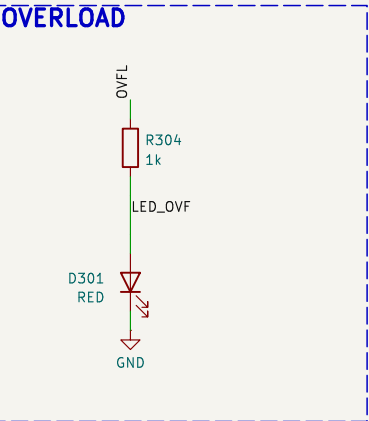
PCM1804 master, 192 kHz



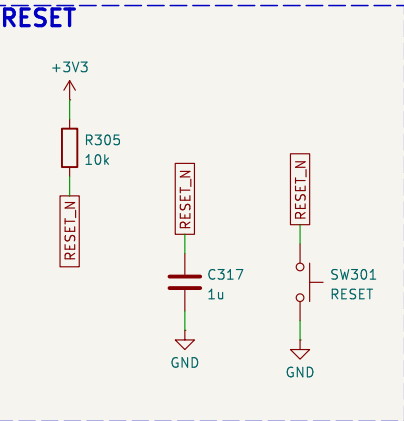
SUPPLY DECOUPLING



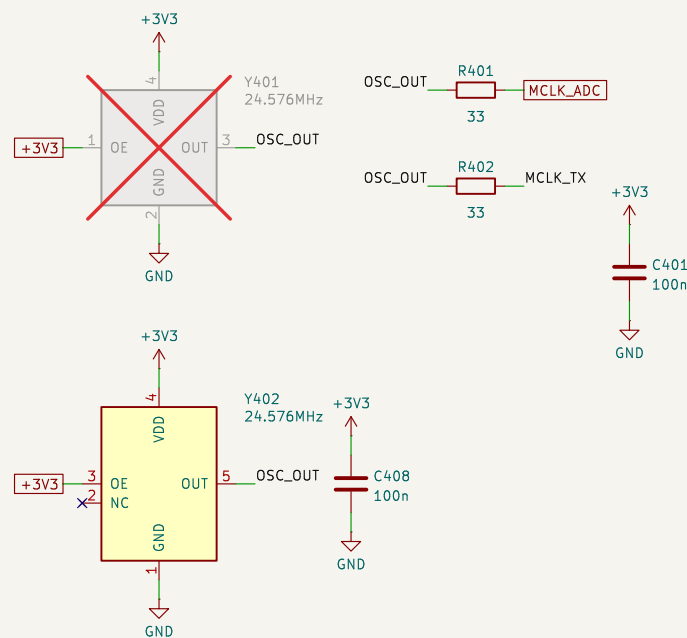
OVERLOAD



RESET

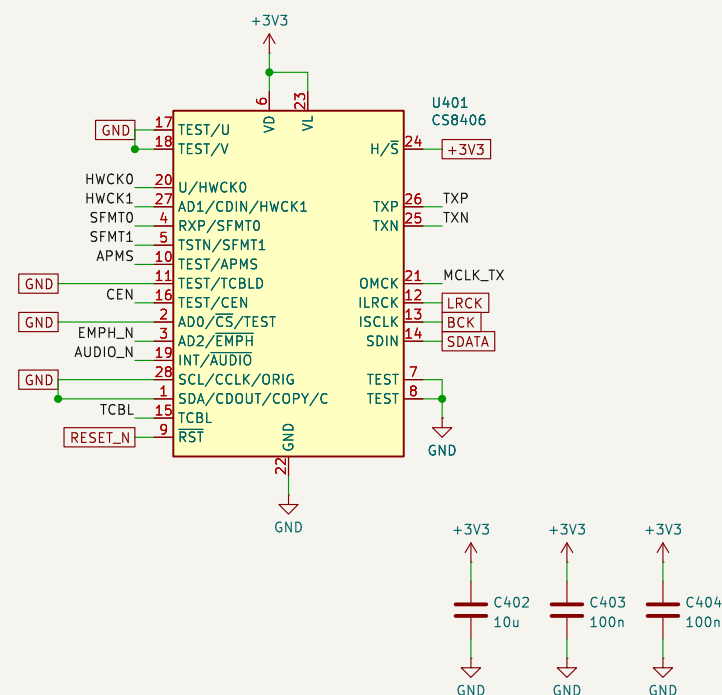


Default (192 kHz): S/M=1 master, OSR2..0=111 (SCKI=128 fs), FMT1..0=01 I2S, BYPAS=1 (HPF off).
Change switches with power off, or press RESET afterwards.

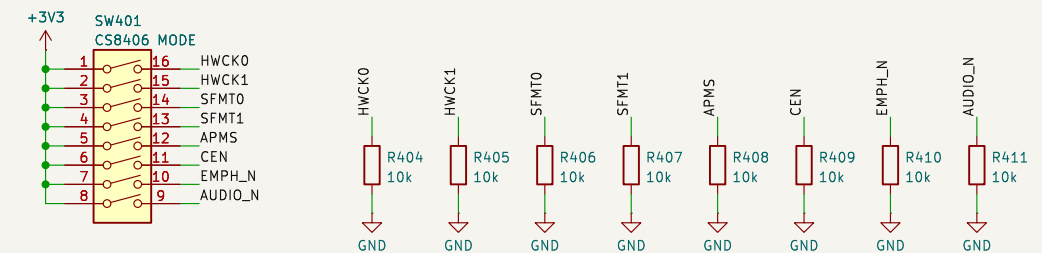


Fit Y402 (SOT23-5, leaded) or Y401 (3.2x2.5 no-lead), never both.

CS8406 hardware mode



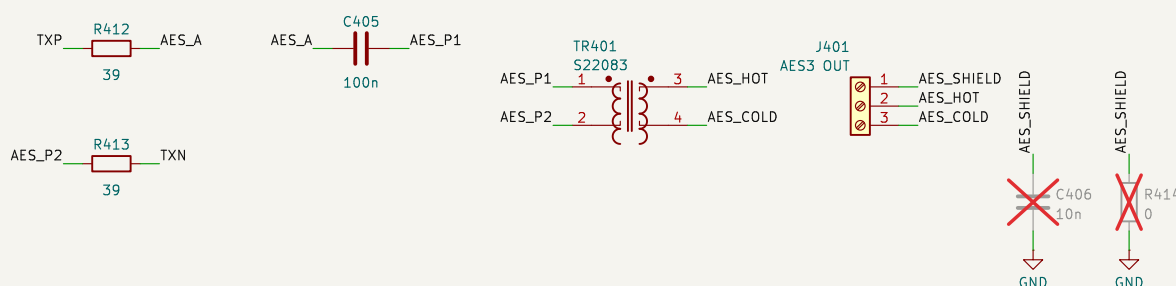
CS8406 MODE (ON = 1)



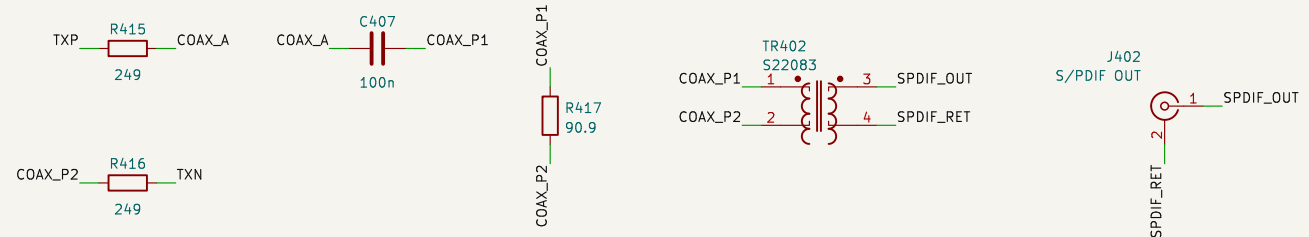
Default (192 kHz): HWCK $\rightarrow$ OMCK = 128 fs, SFMT = 125, APMS=0 (slave), CEN=0, EMPH_N=1, AUDIO_N=0.

OUTPUTS transformer isolated

AES3 110 R balanced



S/PDIF 75 R coax, 0.5 Vpp



AES3: 2 x 39 R + ~2 x 26 R driver ~ 110 R source, ~3 Vpp into 110 R.

S/PDIF: 2 x 249 R + 90.9 R shunt: 75 R source, 0.5 Vpp into 75 R.

